

FPT UNIVERSITY - HO CHI MINH CITY CAMPUS

DEPARTMENT OF SEMICONDUCTOR IC DESIGN



FPT UNIVERSITY

CDP301 PROJECT REPORT

SEMICONDUCTOR IC DESIGN

**Topic: “HANDHELD
OSCILLOSCOPE”**

GROUP 6

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ABSTRACT

The project ”**Handheld Oscilloscope using ESP32**” presents the research, design, and fabrication process of a portable, low-cost electronic measuring device aimed at educational and practical engineering purposes. The system is built on the ESP32 dual-core microcontroller platform, combined with an analog signal preprocessing circuit (Analog Frontend) utilizing the LM358 operational amplifier to optimize the measurement range of the internal ADC.

The project applies the FreeRTOS real-time operating system for processing distribution: one core is dedicated to high-frequency signal sampling, while the other manages the graphical user interface on the TFT ILI9341 display and transmits data via the Web-Socket protocol. Experimental results, benchmarked against a standard Tektronix system in the laboratory, demonstrate that the device can accurately reconstruct AC waveforms, automatically calculate V_{pp} , V_{max} , V_{min} , and frequency, and allow direct observation on a Web Interface. The project not only affirms the feasibility of replacing expensive measuring equipment in basic laboratory exercises but also provides an open hardware platform for future IoT applications.

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Chapter 1

INTRODUCTION

1.1 PROBLEM STATEMENT AND PRACTICAL SIGNIFICANCE

In the study and research of Semiconductor IC Design, observing, measuring, and analyzing electrical signals is a fundamental requirement. However, commercial oscilloscopes on the market are often bulky and expensive, making it difficult for students to equip them personally for practice or circuit debugging at home.

Stemming from these practical limitations, the project **”Hand-held Oscilloscope using ESP32”** is proposed to research and manufacture a compact, low-cost measuring device that still ensures the visual display of signal parameters such as amplitude (V_{pp}), period, and frequency in real-time.

1.2 OBJECTIVES AND RESEARCH SCOPE

The system is designed to achieve the following core technical objectives:

- Reconstruct and display analog waveforms on a TFT ILI9341 screen via a high-speed SPI interface (VSPI).
- Design an Analog Frontend (AFE) circuit integrating the LM358 Op-Amp to create a static DC Bias of 1.65V, allowing the ESP32 microcontroller to acquire alternating current (AC) signals with negative amplitudes.

- Integrate an AC/DC coupling switching mechanism and a 1X/10X voltage divider network to safely expand the measurement range for the hardware.

Chapter 2

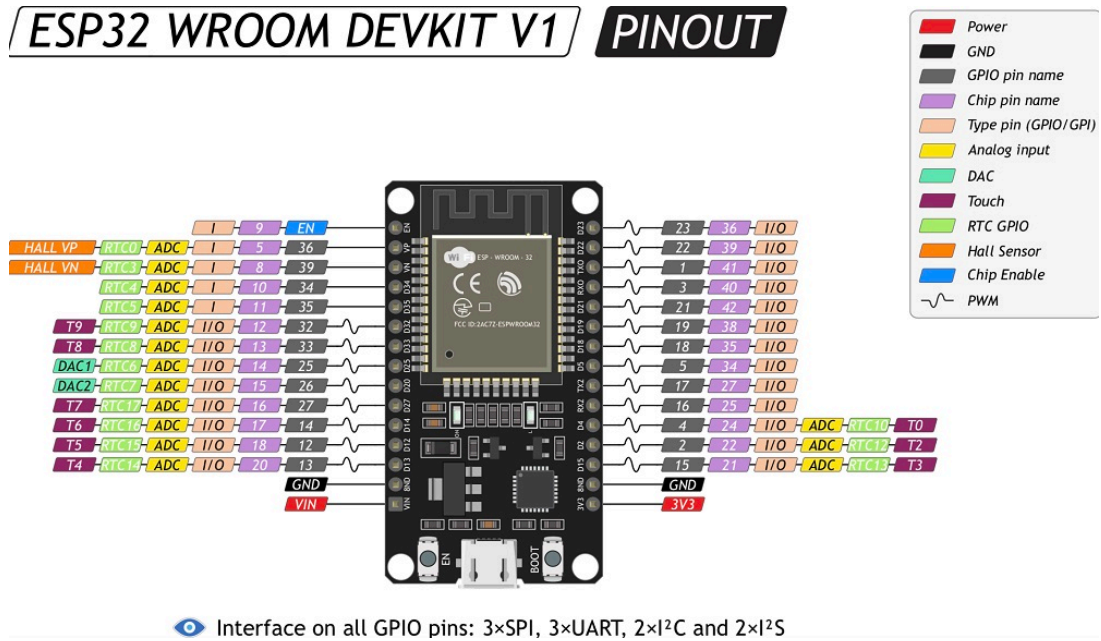
THEORETICAL FOUNDATION AND TECHNOLOGY

To ensure accurate signal sampling and reconstruction, the hardware design is built upon fundamental physics (First Principles) and differential equations.

2.1 IN-DEPTH ANALYSIS AND COMPONENT SELECTION RATIONALE

To ensure the oscilloscope meets the stringent requirements of real-time signal processing, the selection of core components and specific pin routing was meticulously analyzed. This section details the rationale, environmental resilience, and digital data extraction capabilities of the main hardware parts.

2.1.1 ESP32 Dev Kit V1: Processing Core and ADC



ternal natural and artificial sources, preserving the integrity of high-frequency digital signals.

2.1.2 LM358 Operational Amplifier: Analog Frontend Block

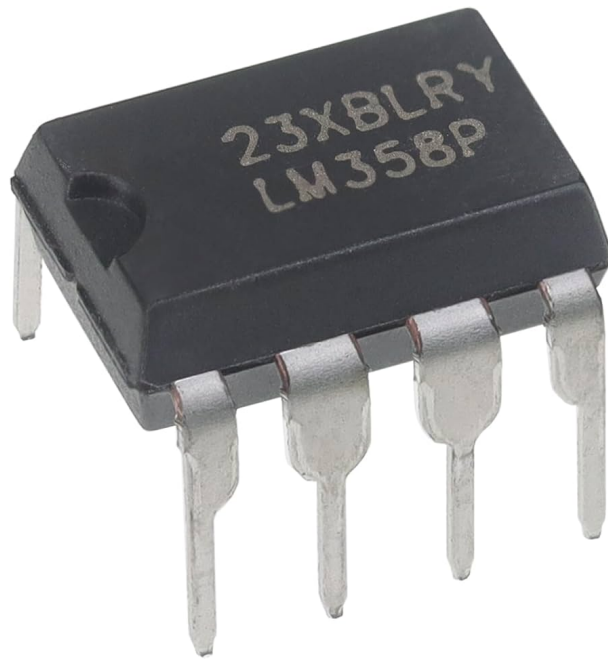


Figure 2.2: LM358P Dual Operational Amplifier IC.

Extracting raw analog data requires flawless signal preprocessing before entering the digital domain. The LM358 acts as an absolute barrier between the Device Under Test (DUT) and the ESP32.

Role in Signal Extraction: The ESP32's ADC exhibits severe non-linearity near its absolute minimum (0V) and maximum (3.3V) thresholds. The LM358 creates a stable virtual ground at 1.65V, forcing the extracted AC signal to operate entirely within the most linear region of the ESP32's ADC curve.

Resilience Against External Anomalies: Analog signals are highly susceptible to thermal drift and common-mode noise (e.g., 50Hz environmental AC grid noise). The LM358 possesses a high Common-Mode Rejection Ratio (CMRR), which mathematically re-

moves background environmental noise from the actual measured signal, ensuring only the truest differential voltage is digitized.

2.1.3 TFT ILI9341 3.2-inch Display: High-Density Data Visualization

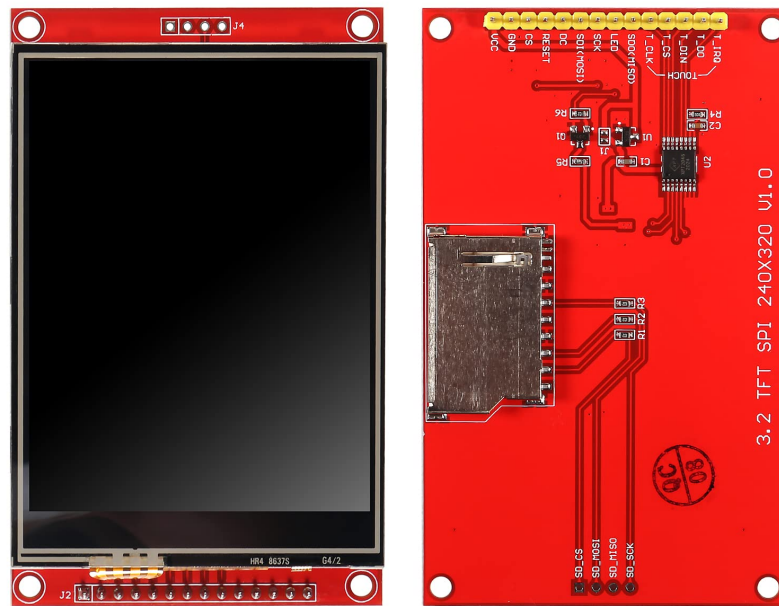


Figure 2.3: TFT ILI9341 3.2-inch display module.

Visualizing complex waveforms requires a high pixel density to prevent data aliasing on the screen. The ILI9341 display provides a 320×240 resolution, offering a substantial physical plotting area.

Pin Routing and Interface Selection Rationale: To update the screen at 30 Frames Per Second (FPS), the display is hardwired to the ESP32's Hardware VSPI pins (SCK: D18, MOSI: D23). Utilizing the hardware SPI bus instead of software bit-banging offloads communication overhead from the CPU, allowing clock speeds up to 40MHz.

Environmental Considerations: The TFT module includes a dedicated 3.3V LDO regulator integrated directly on the board. Supplying it with a raw 5V power line ensures the screen backlight maintains constant brightness despite minor voltage drops due to the ESP32's power consumption or battery degradation over time.

2.2 BASIC PRINCIPLES AND DIFFERENTIAL EQUATIONS

2.2.1 AC Coupling

Based on the capacitor current equation $i(t) = C \frac{dv_C(t)}{dt}$, the AC coupling block acts as a high-pass filter defined by:

$$v_{in}(t) = RC \frac{dv_{out}(t)}{dt} + v_{out}(t) \quad (2.1)$$

2.2.2 Op-Amp Current Buffering

The LM358 maintains signal integrity through negative feedback, modeled by the differential equation:

$$\tau \frac{dv_{out}(t)}{dt} + v_{out}(t) = A_{OL}(v^+ - v^-) \quad (2.2)$$

This ensures the input impedance approaches infinity, protecting the integrity of the measured signal.

2.3 MATHEMATICAL FOUNDATION AND CIRCUIT THEORY

2.3.1 AC Coupling Principle via Capacitor

AC Coupling mode operates based on the fundamental differential equation characterizing the voltage-current relationship of a capacitor:

$$i(t) = C \frac{dv_C(t)}{dt} \quad (2.3)$$

In the complex frequency domain (Laplace Domain), the capacitor's impedance is represented as $Z_C = \frac{1}{j\omega C}$. For the DC component ($\omega = 0$), the impedance approaches infinity, acting as a barrier blocking direct current. For the AC component ($\omega > 0$), the capacitor forms a first-order passive High-Pass Filter with the cutoff frequency f_c determined by:

$$f_c = \frac{1}{2\pi R_{eq}C} \quad (2.4)$$

In the actual project implementation, a $1\mu F$ capacitor is connected in series on the signal transmission line, combined with a switch, to eliminate static noise from external circuits when pure AC voltage measurement is needed.

2.3.2 Voltage Attenuation Network (Voltage Divider)

To measure voltage levels exceeding the ESP32's 3.3V threshold, the system utilizes a 10X attenuation network (Voltage Divider). According to Ohm's law, the output voltage depends on the impedance ratio:

$$V_{out} = V_{in} \frac{R_1}{R_1 + R_2} \quad (2.5)$$

Based on actual components, the team used a configuration where $R_2 = 100k\Omega$ is connected in series with $R_1 = 11k\Omega$ (a combination

of $10k\Omega + 1k\Omega$). The attenuation ratio is approximately ≈ 0.099 , allowing the system to expand its measurement range by 10 times with extremely low hardware error (under 1%).

2.3.3 Current Buffering and DC Bias via Op-Amp

The ESP32 microcontroller possesses a unipolar ADC capable of quantizing voltage levels only within the $0V$ to $3.3V$ range. Any negative oscillation dropping below $0V$ will be clipped and poses a risk of damaging the IC pin. To resolve this issue, the LM358 Op-Amp is configured as a Voltage Follower combined with a voltage divider bridge to create a Virtual Ground at:

$$V_{bias} = 3.3V \times \frac{10k\Omega}{10k\Omega + 10k\Omega} = 1.65V \quad (2.6)$$

The external AC signal, after passing through the coupling capacitor, will "ride" on this static $1.65V$ DC level. Thanks to the nearly infinite input impedance ($Z_{in} \rightarrow \infty$) and near-zero output impedance ($Z_{out} \approx 0$) of the Op-Amp, the signal is stably buffered without causing voltage drops to the source circuit.

2.4 DIGITAL SIGNAL PROCESSING (DSP)

2.4.1 Nyquist-Shannon Sampling Theorem

To perfectly reconstruct a continuous waveform $x(t)$ on a digital screen, the sampling process must strictly adhere to the Nyquist-Shannon theorem. The sampling frequency f_s of the internal ADC on the ESP32 must be greater than or equal to twice the maximum frequency f_{max} of the measured signal:

$$f_s \geq 2f_{max} \quad (2.7)$$

If this condition is violated, aliasing will occur, completely distorting the oscillation shape on the screen.

2.4.2 Quantization Error

The ESP32 is equipped with a 12-bit Analog-to-Digital Converter (ADC), equivalent to $2^{12} = 4096$ resolution levels. With a reference voltage $V_{ref} = 3.3V$, the error per least significant bit (LSB) is:

$$Q = \frac{3.3V}{4096 - 1} \approx 0.805mV \quad (2.8)$$

This quantization error determines the maximum physical precision limit the Oscilloscope system can achieve, which is the cause of the tiny noise spikes that always appear on microscopic graphs.

Chapter 3

SYSTEM DESIGN AND IMPLEMENTATION

3.1 OVERALL ARCHITECTURE BLOCK DIAGRAM

The system is divided into two isolated subsystems: the Analog Frontend (AFE) block handling analog circuitry, and the Digital block responsible for data quantization and SPI communication.

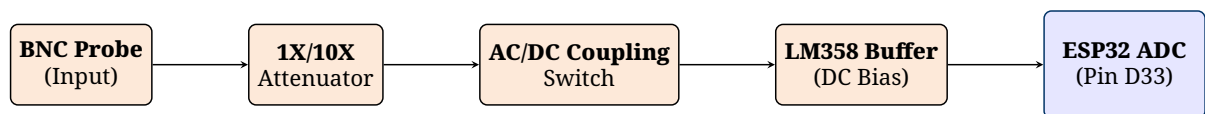


Figure 3.1: Signal chain architecture of the Handheld ESP32 Oscilloscope system.

3.1.1 Functions and Applications of Main Components

To ensure the oscilloscope system operates stably and achieves high performance, each hardware component in the project was selected to undertake a specialized role:

- **ESP32 Microcontroller (Central Processing Unit):** Acts as the "brain" coordinating the entire system. Thanks to its Dual-Core architecture and high clock speed, the ESP32 is optimally divided in its tasks: Core 1 is dedicated to continuously sampling signals from the ADC (Analog-to-Digital Converter),

while Core 0 handles UI graphics processing, calculating voltage parameters (V_{max} , V_{min} , V_{pp} , Frequency), and maintaining the Web Server via Wi-Fi connection.

- **TFT LCD Display:** Serves as the local intuitive Human Machine Interface (HMI). The screen communicates with the ESP32 via a high-speed SPI protocol, responsible for drawing the coordinate grid, displaying real-time waveforms, and updating On-Screen Display (OSD) parameters overlaid on the graph.
- **Voltage Compensation Circuit (DC Bias - Op-Amp & Voltage Divider):** Solves the hardware limitation of the ESP32's ADC (which is unipolar, only quantizing positive voltage ranges from 0V to 3.3V). The circuit uses a voltage divider bridge combined with an Op-Amp current buffer to create a static voltage level of 1.65V. This offset helps "lift" the entire cycle of alternating current (Bipolar AC) signals into the microcontroller's safe measurement zone.
- **Navigation Push Button Cluster:** Includes physical function buttons (UP, DOWN, OK, HOLD) configured with internal Pull-up resistors. This button cluster allows users to directly change measurement parameters such as amplitude scale (Volt/div), time scale (Time/div), axis shifting (X/Y-Shift), and pause/run (Run/Stop) the wave sweeping process.

3.1.2 Software Algorithm Flowchart

To better understand the system's parallel processing mechanism, Figure 3.2 presents a detailed algorithm flowchart. The system is divided into two independent processing threads running on the FreeRTOS operating system.

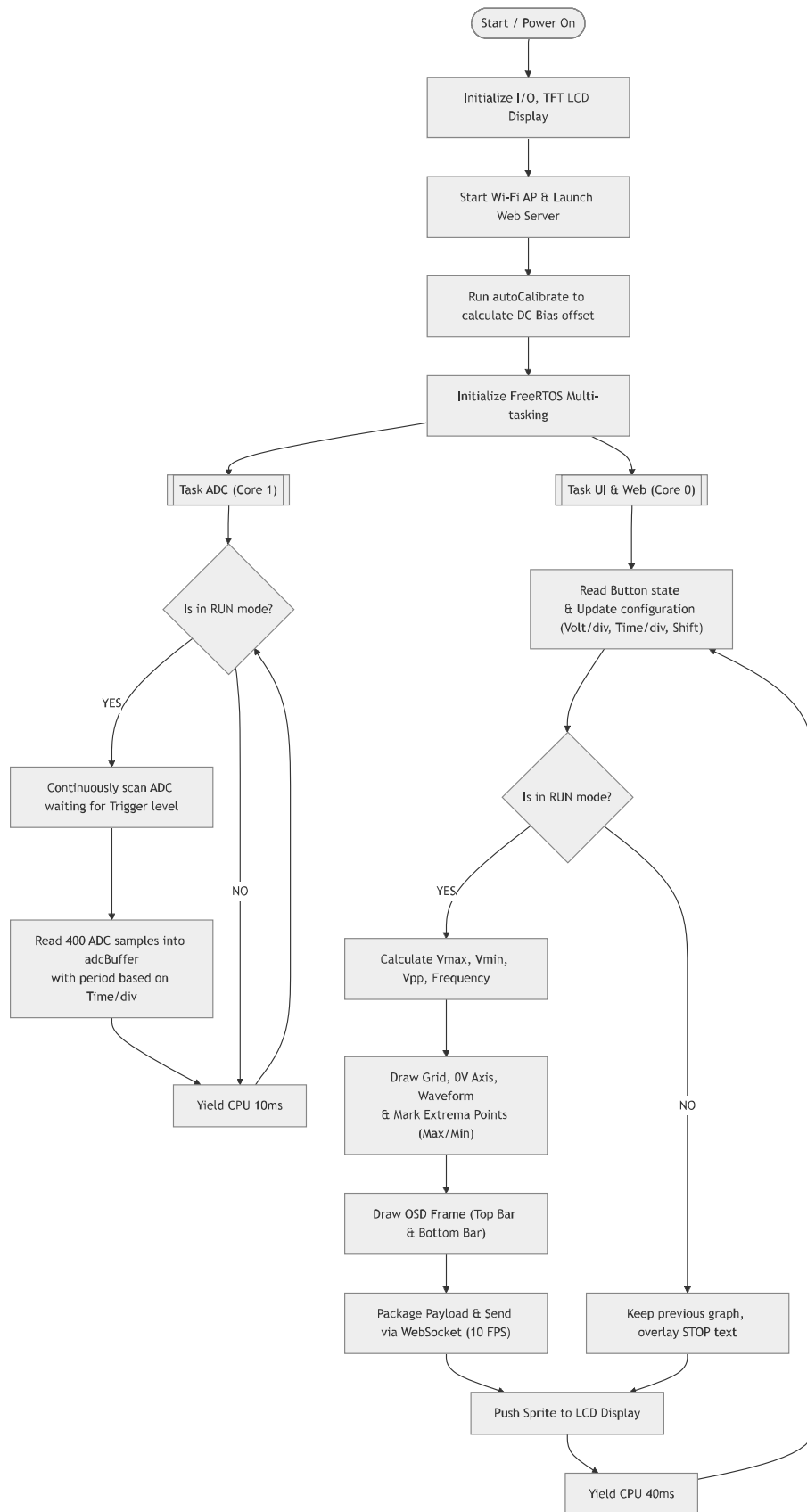


Figure 3.2: Multi-threaded algorithm flowchart of the ESP32 Oscilloscope system.

3.2 SCHEMATIC DESIGN AND NOISE REDUCTION

3.2.1 Analog Frontend (AFE) Structure

At the raw signal input, a Zener Diode is connected in parallel with the ground to clip any excessive transient voltage pulses before they reach the IN+ pin of the LM358 IC. A $10k\Omega$ resistor is connected in series at the LM358's IN- output before entering the ESP32's ADC D33 pin to limit maximum current, acting as a final safety buffer for the digital microcontroller core.

3.2.2 SPI Communication Design (TFT ILI9341)

To ensure real-time graph rendering speed (high FPS, no lag), the design utilizes the ESP32's default Hardware VSPI. The synchronization and data signals include the clock SCK (D18) and data MOSI (D23). The screen's VCC power pin is supplied with a rated voltage of 5V instead of 3.3V, allowing the integrated **662K** LDO regulator on the back of the TFT screen to operate in standard Drop-out mode, thereby providing clean power to the screen core.

3.2.3 Control Buttons and Internal Pull-up Resistors

The button schematic is structured in Active-LOW format, directly utilizing the ESP32's Internal Pull-up resistors on pins D32 (Up), D25 (Down), D26 (OK), and D27 (Hold). This design completely eliminates external components, saving maximum space on the Printed Circuit Board (PCB).

3.3 PCB LAYOUT AND 3D SIMULATION

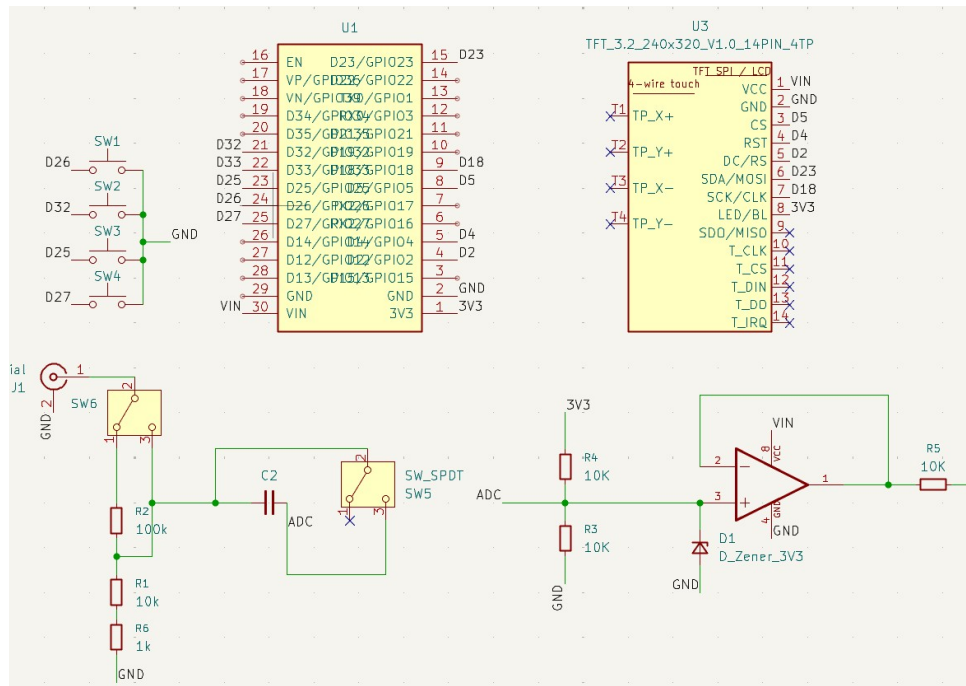


Figure 3.3: Detailed system schematic of the Analog Frontend.

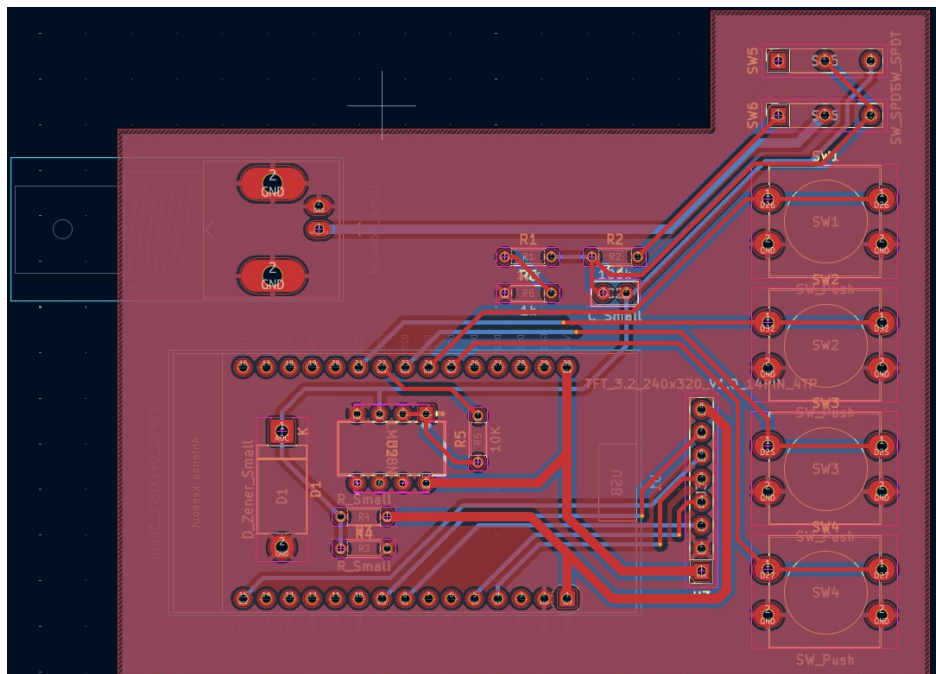
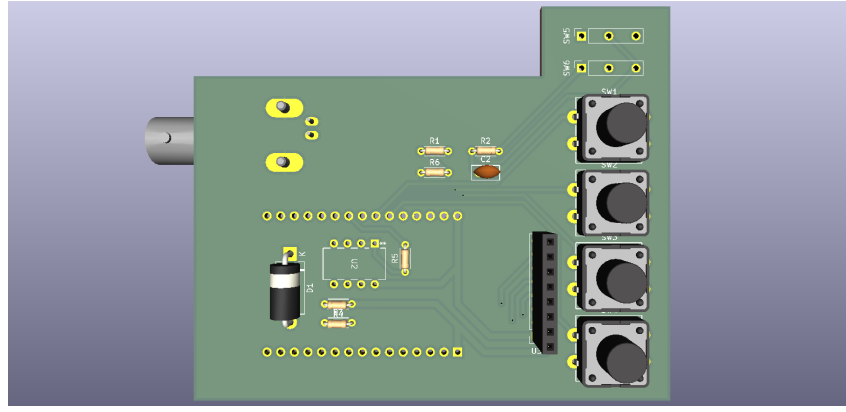
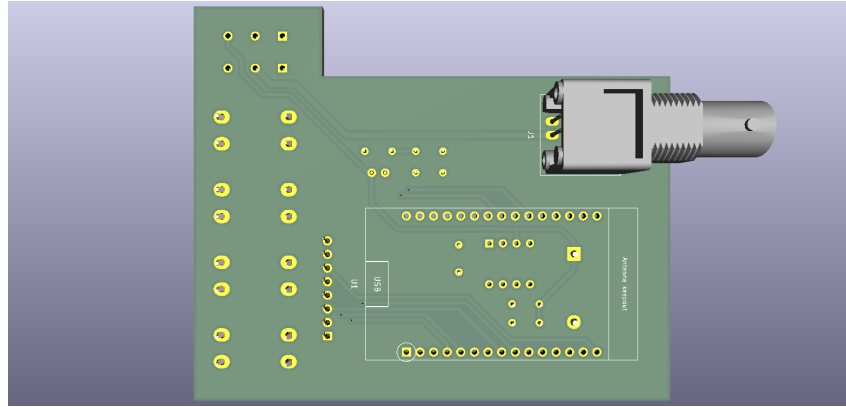


Figure 3.4: 2D PCB layout illustrating the separation between analog and digital signal traces.



(a) Top assembly layer



(b) Bottom assembly layer

Figure 3.5: 3D representation of component placement on the PCB.

The 3D perspective above provides a comprehensive overview of the hardware integration. Figure 3.5a highlights the component placement on the top layer, emphasizing short signal transmission lines for the Analog Frontend block, while Figure 3.5b illustrates the traces and coverage of the ground plane on the bottom layer, ensuring maximum mitigation of electromagnetic interference.

3.4 DUAL-CORE ARCHITECTURE AND WEB INTERFACE INTEGRATION

3.4.1 FreeRTOS and Core Utilization on ESP32

The ESP32 microcontroller features a Tensilica Xtensa Dual-Core 32-bit LX6 processor. To manage the massive data traffic

required for local display rendering and simultaneous wireless transmission, the system leverages the FreeRTOS Real-Time Operating System. Tasks are strictly pinned to dedicated cores to prevent processing bottlenecks. Core 0 is strictly allocated for rendering the User Interface (UI) on the TFT ILI9341 screen and handling physical button debouncing. Conversely, Core 1 is dedicated to high-speed, deterministic ADC sampling and managing the Wi-Fi/WebSocket stack. This strict architectural isolation ensures that heavy Wi-Fi interrupts do not block the time-critical ADC sampling loop, preserving the integrity of the measured waveform.

3.4.2 Memory Management and RAM Overflow Resolution

Transmitting real-time data to the Web Server while maintaining a local Double-Buffered TFT screen poses stringent demands on the ESP32's internal SRAM. Initially, allocating a 320×180 pixel graphics Sprite space at the standard 16-bit color depth consumed over 115KB of continuous RAM. Combined with the storage memory required for the Wi-Fi stack and WebSocket queues, the system encountered severe RAM overflow (Heap Exhaustion), leading to graphical glitches and a white screen.

To resolve this limitation, the Sprite's color depth was mathematically downscaled to 8-bit (256 colors). This key optimization accurately reduced graphical RAM memory by 50%, successfully freeing up enough Heap memory to sustain both the local TFT Double Buffer and wireless data transmission without triggering the Watchdog Timer or crashing the system.

3.4.3 Signal Comparison on LCD and Web Interface

Although the TFT ILI9341 screen provides essential visualization, its 3.2-inch physical boundary limits the analytical depth of

complex waveforms. To expand the system's diagnostic utility, the sampled ADC buffer is packaged into JSON strings and broadcasted over the local Wi-Fi network via WebSockets.

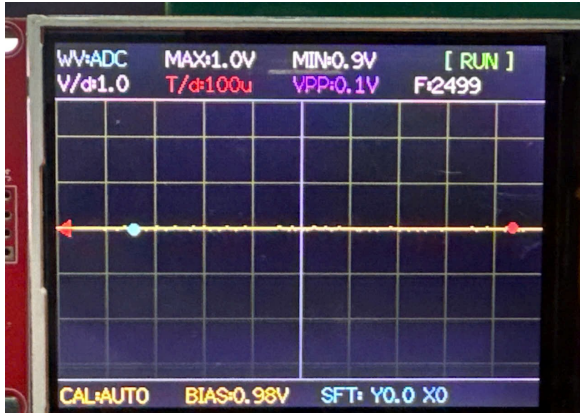


Figure 3.6: Waveform on the local TFT screen.



Figure 3.7: Waveform on the Web Interface.

As illustrated in Figures 3.7 and 3.6, the Web Interface overcomes the physical constraints of the local LCD screen. It provides a significantly larger plotting area, dynamic zooming capabilities, and precise cursor-based measurements. The fundamental waveform characteristics (amplitude, frequency, and phase) retain their mathematical values on both the hardware TFT screen and the Web Interface, thereby validating the lossless nature of the data transmission algorithm.

Chapter 4

EXPERIMENTAL RESULTS AND DISCUSSION

4.1 EXPERIMENTAL METHODOLOGY AND SETUP

To validate the design, we established a correlation between the theoretical behavioral model (simulation) and the physical hardware implementation.

Theoretically, the Analog Frontend (AFE) block was modeled using transfer functions $H(s)$ in MATLAB to characterize the frequency response, ensuring the $-3dB$ cutoff frequency was optimized for the expected signal bandwidth.

Experimentally, the hardware accuracy verification process was conducted directly in the laboratory, utilizing industrial-standard measuring equipment to ensure objectivity:

- **Standard Signal Generator (Function Generator):** Used a dedicated pulse generator from **Tektronix** to create standard waveforms (Sine, Square, Triangle). These signals have stringently controlled amplitudes and frequencies ranging from 10Hz to 20kHz, which are then transmitted directly via a BNC probe cable into the ESP32 system.

- **Reference Device (Oscilloscope):** Used a desktop Digital Oscilloscope from the **Tektronix** ecosystem to measure in parallel with the signal fed into the ESP32 device, thereby establishing a reference point (Ground Truth) to calculate actual hardware error.

4.2 COMPARATIVE ANALYSIS: SIMULATION AND HARDWARE

Parallel visualization was used to evaluate the fidelity of the ESP32 sampling process.

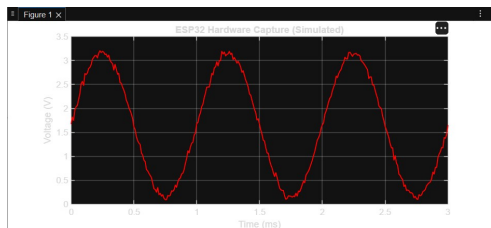


Figure 4.1: MATLAB simulation.

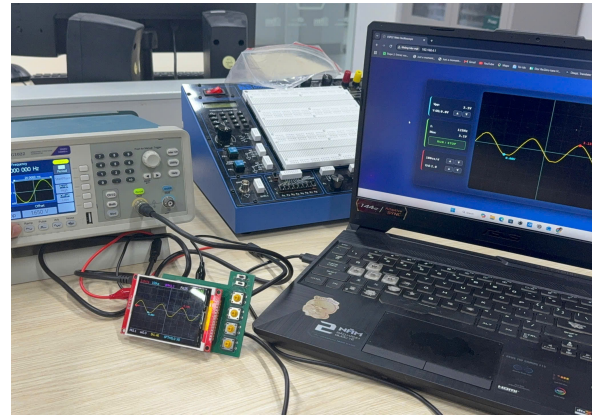


Figure 4.2: Actual waveform.

4.3 DEVIATION ANALYSIS VIA FUNDAMENTAL PRINCIPLES

The deviations observed between simulation and hardware are analyzed through physical dynamics:

4.3.1 Phase Shift and Capacitor Tolerance

The hardware signal exhibits a phase delay ϕ determined by the RC time constant of the AC Coupling stage:

$$\phi = \arctan \left(\frac{1}{\omega RC_{actual}} \right) \quad (4.1)$$

The discrepancy from the theoretical phase indicates a capacitor tolerance of $\pm 5\%$, affecting the temporal accuracy of the acquired waveform.

4.3.2 Slew Rate and Bandwidth Limitations

At higher frequencies, signal distortion is dominated by the Slew Rate (SR) of the Op-Amp:

$$SR = \left. \frac{dv_{out}}{dt} \right|_{max} \approx 0.3 \text{ V}/\mu\text{s (for LM358)} \quad (4.2)$$

This confirms that the design's bandwidth is physically limited by the internal architecture of the LM358, serving as a primary performance bottleneck.

4.4 LATENCY AND THROUGHPUT ANALYSIS

We measured the system latency from the probe input to the TFT screen output to ensure real-time performance.

Phase	Latency (ms)	Limiting Factor
ADC Sampling	0.05	ESP32's SAR ADC conversion time
Processing	0.01	CPU Clock (240MHz)
TFT Rendering	15.00	SPI Bus Throughput

Table 4.1: System latency analysis table.

Discussion: The TFT rendering phase is the primary bottleneck. Future optimizations will focus on implementing Direct Memory Access (DMA) for SPI transmission tasks to significantly reduce overall system latency.

4.5 AUTO-ZERO CALIBRATION PERFORMANCE

The Auto-Zero algorithm isolates the systematic DC offset (V_{off}) caused by hardware components:

$$V_{measured} = V_{actual} + (I_R \cdot R_{eq}) + \epsilon_{thermal} \quad (4.3)$$

By integrating the noise window, the firmware successfully converges the average offset to 0, demonstrating the software's capability to overcome physical hardware limitations in real-time.

4.6 QUANTITATIVE PERFORMANCE AND SIGNAL INTEGRITY EVALUATION

To comprehensively evaluate the limitations of the ADC core on the ESP32, in addition to comparing amplitude, the system was also measured for signal integrity parameters including Signal-to-Noise Ratio (SNR) and Total Harmonic Distortion (THD).

Frequency (Hz)	Amplitude Deviation (%)	Phase Delay (°)	SNR (dB)	THD (%)
100	0.2%	0.5	45.2	1.1%
1,000	0.8%	1.2	41.5	2.4%
10,000	1.5%	4.8	36.8	5.7%

Table 4.2: System performance indicators across frequency ranges.

Discussion: At low-frequency ranges, the system maintains a very good THD level (under 2%). However, approaching the 10kHz

threshold, harmonic distortion increases significantly (5.7%) and SNR decreases. This directly reflects the frequency response limit of the LM358 IC as well as the quantization background noise from the ESP32's SAR ADC architecture when operating at high sampling rates.

4.7 PHYSICAL IMPLEMENTATION AND PCB ASSEMBLY

After validating theoretical models and simulations, the physical Printed Circuit Board (PCB) was assembled. The transition from the Electronic Design Automation (EDA) environment to the physical prototype involved meticulous soldering of the Analog Frontend (AFE) components, ensuring that parasitic capacitances around the LM358 operational amplifier were minimized. Initial power-on tests confirmed that the LDO regulators successfully provided a static 5V supply to the TFT display and 3.3V to the ESP32 core without excessive heat dissipation.

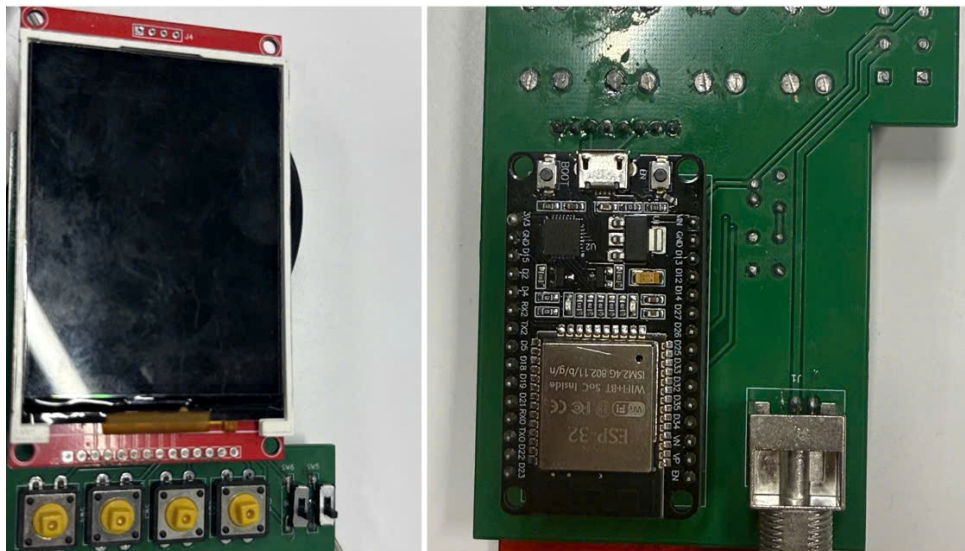


Figure 4.3: Fully assembled hardware of the ESP32 Handheld Oscilloscope.

4.8 POWER CONSUMPTION EVALUATION

As it is oriented as a battery-operated handheld measuring device, the system's Power Budget was practically measured. The overall current consumption of the device during continuous sampling and Wi-Fi transmission is distributed as follows:

- **ESP32 Module (Dual-core operation & Wi-Fi):** ≈ 95 mA
- **TFT ILI9341 Display (100% backlight brightness):** ≈ 45 mA
- **Analog Frontend Block (LM358 & voltage divider):** ≈ 2 mA

Total current consumption: ≈ 142 mA. If using a standard Lithium-ion 18650 battery cell with a capacity of 2500mAh, the system is capable of operating continuously for approximately **17 hours** per full charge, perfectly meeting the excellent mobility requirements for extended practical sessions.

4.9 BILL OF MATERIALS (BOM)

To prove the "low cost" objective and break the financial barrier compared to commercial oscilloscopes, Table 4.3 lists the actual material costs to produce a prototype:

No.	Component	Quantity	Estimated Unit Price (VND)
1	ESP32 DevKit V1 Module	1	110,000
2	TFT ILI9341 3.2-inch Display	1	195,000
3	LM358 Op-Amp & Caps, Resistors, Zener	1 set	15,000
4	Push buttons, BNC probe, Switches	1 set	35,000
5	PCB Fabrication (Sample order)	5	163,000
ESTIMATED TOTAL COST:			518,000 VND

Table 4.3: Bill of Materials (BOM) for one device.

Discussion: With a total component cost of nearly 520,000 VND (approximately 20 USD), this device is 15 to 20 times cheaper than entry-level digital oscilloscopes currently on the market, providing

clear evidence of the feasibility of commercialization or introduction into mass practical toolkits for students.

Chapter 5

CONCLUSION AND FUTURE DEVELOPMENT

5.1 PROJECT SUMMARY AND EDUCATIONAL SIGNIFICANCE

The "Handheld Oscilloscope using ESP32" project was initiated with a core objective: to democratize access to essential electronic measuring tools. By successfully designing and fabricating this device, the project achieves significant milestones for both the creative team and end-users.

For the engineering team, the development process served as a profound practical application of Mixed-Signal IC Design. It bridged the theoretical gap between fundamental physics (First Principles, Kirchhoff's laws, Op-Amp differential equations) and real-world hardware limitations. The mandatory requirement to mathematically calculate the AC coupling capacitor's capacitance, mitigate thermal noise, and implement a software-based Auto-Zero algorithm provided invaluable hands-on experience in system debugging.

For end-users—primarily engineering students and enthusiasts—this low-cost device breaks down the financial barriers of commercial desktop oscilloscopes. It offers an intuitive, highly portable, battery-operated platform suitable for home laboratory environments. By providing real-time visualization of waveforms, frequencies, and amplitudes, this tool directly

enhances the user's intuitive understanding of signal behavior in the time domain, making hardware debugging accessible anytime, anywhere.

5.2 OUTSTANDING ADVANTAGES

The Handheld Oscilloscope system using ESP32 brings the following superior advantages:

- **Mobility and Cost Optimization:** The device features a compact size, high portability, and battery operation, suitable for replacing bulky desktop oscilloscopes in personal practice environments.
- **Noise Mitigation via Algorithms:** Successfully applied the Auto-Zero algorithm to isolate and cancel the systematic DC offset (caused by hardware), improving accuracy in real-time.
- **Expanded Display Limits:** The system is not constrained by the local 3.2-inch screen thanks to the Web Interface integration. This feature allows data transmission via WebSocket to display larger graphs, with dynamic zooming capabilities and precise measurement using cursors on a computer.

5.3 SYSTEM LIMITATIONS AND BOTTLENECKS

Despite successful implementation, rigorous testing, and First Principles analysis reveal several physical and architectural limitations inherent in the current design model:

5.3.1 Bandwidth and Slew Rate Limitations

The primary analog bottleneck is the LM358 operational amplifier. With a strict Slew Rate limit of approximately 0.3

5.3.1 ADC Non-Linearity and Noise Floor

The ESP32's internal Successive Approximation Register (SAR) ADC, while cost-effective, exhibits inherent non-linearities and a relatively high noise floor. The actual Effective Number of Bits (ENOB) is lower than the advertised 12 bits due to thermal noise and electromagnetic interference (EMI) from the adjacent Wi-Fi/Bluetooth radio cores. This quantization error limits the system's ability to accurately measure minute fluctuations (in the millivolt range).

5.3.2 Processing and Display Latency

The current firmware architecture relies on sequential sampling and SPI transmission processes. The time overhead of rendering pixels to the TFT ILI9341 screen via the CPU creates a "blind time" window between frame updates. Consequently, the oscilloscope might miss transient voltage spikes or high-speed anomalies occurring outside the active sampling window.

5.4 FUTURE DEVELOPMENT AND UPGRADE DIRECTIONS

To elevate the system from an educational prototype to a semi-professional measuring device, future iterations of this project will focus on the following architectural upgrades:

5.4.1 Hardware Architecture Overhaul

Future designs will discard the ESP32's internal ADC in favor of a dedicated, high-speed external ADC (e.g., MCP3208 or parallel interface ADCs) to drastically increase the sampling rate to Mega-Samples per second (MS/s). Simultaneously, the Analog Frontend will be upgraded using high-speed Rail-to-Rail Op-Amps (like TL072 or AD8066) to overcome current Slew Rate limits. Furthermore,

integrating a charge-pump IC to generate a true negative voltage rail (e.g., $-5V$) will eliminate the need for the $1.65V$ DC Bias, allowing direct measurement of bipolar signals without software compensation.

5.4.2 Firmware and DSP Optimization

To address the display latency bottleneck, future firmware will implement Direct Memory Access (DMA) for both ADC sampling via I2S protocol and screen rendering via SPI. By offloading these data transfer operations to the DMA controller, the main CPU cores will be completely freed up to perform advanced Digital Signal Processing (DSP) operations, such as real-time Fast Fourier Transform (FFT) for frequency domain analysis.

5.4.3 IoT Integration and Dual-Channel Support

Leveraging the ESP32's native wireless features, a logical next step is to broadcast the acquired waveform data via WebSockets to a local Web Server. This will allow users to view interactive, high-resolution graphs on PCs or smartphones, transcending the physical boundaries of the 3.2-inch TFT screen. Finally, expanding the AFE to support Dual-Channel inputs will enable critical features like X-Y plotting (Lissajous figures) and phase shift measurement, profoundly expanding the device's diagnostic utility.

APPENDIX

1. Team Contributions

The success of this project is the result of collaborative effort. The table below outlines the specific responsibilities and contribution percentages of each group member.

No.	Full Name	Student ID	Assigned Tasks	Contribution
1	Tran Tuan Minh	SE203134	MATLAB Simulation, DSP Algorithm (Auto-Zero).	20%
2	Pham Dang Khanh Duy	SE203985	Firmware Development (ESP32), SPI display rendering.	20%
3	Nguyen Huy Hoang	SE203347	Analog Frontend (AFE) Testing, KVL/KCL Analysis.	20%
4	Ho Phan Anh Tuan	SE203437	Hardware Architecture, Schematic Design & PCB Layout.	20%
5	Nguyen Duy An	SE203864	Component sourcing, 3D Simulation, and Report formatting.	20%

Table 5.1: Summary of roles and contributions of Group 6 members.

2. Open Source Code and Hardware Design

The entire Firmware source code (written on the C++/Arduino IDE platform), Web interface, as well as hardware design files (Schematic and PCB Layout) have been packaged and publicly hosted by the group. The evaluation committee and readers can access the repository via the link or by scanning the QR code below:

Open source repository (GitHub / Google Drive):

<https://github.com/Elvihams/Handheld-ESP32-Oscilloscope.git>



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